Assignment - 4

CSL 2050 - Pattern Recognition & Machine Learning

Max. Marks: 50

Submission Instructions:

- Students must submit one Jupyter notebook (.ipynb) file containing:
  - Executable code
  - Visible outputs (tables, figures, metrics)
  - Brief comments or markdown explanations where required
- File naming format : **RollNo.-Name.ipynb** example, 21CS045-AmitSharma.ipynb
- Do Not submit PDFs, word files, or screenshots. ● Late submissions will be penalized as per course policy. ● Date of last submission: **15th April 2026, 11:59PM**.

Part A: Decision Trees [use library]

Dataset: Breast Cancer Wisconsin
https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data/code

Tasks:

1. Perform exploratory analysis i.e., evaluating data distribution, printing feature shape, class distribution. [5]

2. Implement Impurity measures i.e., Gini index and entropy [5]

3. Information Gain: [5] A. For each feature, try possible thresholds and compute impurity reduction
   B. Return best feature and threshold

4. Implement the decision tree (using library) in order to create a generalized classification algorithm (that does not overfit). Visualize the decision tree and its boundary [5]

5. Prediction and Evaluation [5] A. Predict labels for test data
   B. Compute accuracy. Visualize the results via confusion matrix

Part B: Support Vector Machines [use library]

Dataset: Same as above (Breast Cancer Wisconsin)

Tasks:

1. Data Preparation Pipeline: [5] A. Split dataset into train/test sets
   B. Perform feature scaling using StandardScaler
   C. Verify shapes of processed data

2. Train an SVM with linear kernel [use library]. Showcase the number of support vectors, model coefficients (if available). [5]

3. Model Evaluation: [5] A. Predict on test set
   B. Compute accuracy and confusion matrix

4. Kernel Comparison: [5] A. Train SVM with RBF kernel.
   B. Compare the accuracy of the RBF kernel with the linear kernel.

5. Hyperparameter Tuning: [5] A. Experiment with parameters i.e., C (regularization) and gamma (for RBF)
   B. Create a table showing parameter values vs accuracy.